Modern Academy
for Engineering and Technology in Maadi
Electrical Engineering

MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

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Academic Level:	Senior 2

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELCN425) Digital Signal Processing
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan
Branch: Communications & Computers
Time: 2 hours

Number of Pages: 1 | Number of Questions: 4 | Attempt all questions

Question 1 [10 Points] [LO's: 1, 2]

a- Find the relationship between the frequencies, f, F & F_S, where these frequencies respectively are: frequency of analog signals, frequency of discrete-time signals and sampling frequency. 2

b- Compare between FIR and IIR filters. 2

c- Consider the following analog sinusoid signal: $x_a(t) = 2\cos(200\pi t)$

(i) Sketch the signal for [0, 15] ms. 6

(ii) The signal is sampled with sampling rate F_S=300 samples per second, find x(n) and its frequency.

(iii) Compute the sample values of x(n) in one period.

Question 2 [10 Points] [LO's:2, 4]

a- The DFT has a number of mathematical properties which can be used to simplify problems or which lead to useful applications, state some of them (at least 3 properties). 1.5

b- The choice between FIR and IIR filters depends on the relative advantages of the two filter types, how do you choose between these two filters? 1.5

c- For the following sequence: x(n) = {1, 0, 1, 1}:

(i) Using the FFT algorithm, find the frequency components of this sequence. 2.5

(ii) Check your results obtained in part (i) by calculating x(n) using the IDFT method. 2.5

(iii) Using your results, verify Parseval's theorem. 2

Question 3 [10 Points] [LO's: 1, 3]

a- Using Fig. s and Eq. s, define the following items: Correlation operation and End effect problem. 2

b- Using the direct method, calculate the cross-correlation of the following data sequences, x_1 = {1, 1, 0, 1} and x_2 = {1, 0, 0, 1}. Then plot the obtained correlation function against lag number, j. 4

c- For the given waveform shown in Fig. 1, Analytically, compute the convolution with itself, to find the system o/p, y(t). 4

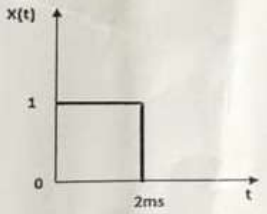


Fig. 1

Question 4 [10 Points] [LO's: 2, 4]

a- For LTIS define the following items: Linearity and Causality. 2

b- For the following transfer function: $H(z) = \dfrac{z^2(z+1)}{(z-0.5)(z-1)^2}$

(i) Plot the pole - zero diagram of H (z). 1

(ii) Find the difference equation of this filter, y (n). 2

(iii) Represent the filtering operation in a block diagram form. 1

c- Using the partial method, find the impulse response, h(n), of the system characterized by H(z) given in part (b). 4

Z-transforms of some required sequences.

Discrete-time sequence x(n), n ≥ 0	z-transform X(z)	Note
kn	$\frac{kz}{(z-1)^2}$	k and α are constants
k	$\frac{kz}{z-1}$	
$k\alpha^n$	$\frac{kz}{z+a}$	

Good Luck